HBK/FJ-2008 Die-Lubricant Spray Research

600-ton hot forging press | Steel ~1200°C | Die Ø250–300 mm

1. Executive conclusion

Recommended first nozzle to test: CY Spraying & Purification Technology **WA-100-10**, 1.0 mm. The published specification gives approximately 190 ml/min liquid output and approximately 250 mm spray pattern at 200 mm distance, making it a practical first trial for a Ø250–300 mm die. This is an engineering starting point, not a guaranteed production setting.

2. Process conditions

Parameter	Condition
Lubricant	HBK/FJ-2008 (also called HBK-500/FJ-2008 during the discussion)
Dilution	1 : 7 lubricant : water — ratio supplied by the user; not independently verified in public TDS
Material	Steel
Forging temperature	~1200°C
Press	600 t
Die diameter	250–300 mm
Application	Die lubrication/release and cooling by spray

3. Lubricant research

Public manufacturer information identifies HBK/FJ-2008 as an environmentally friendly forging die-release lubricant and a polymer water-soluble powder. The public material reviewed did not provide a verifiable manufacturer-published powder-to-water ratio. Therefore, this report uses the user's 1:7 ratio as the working ratio and recommends confirmation against the manufacturer's TDS before production.

Do not substitute another lubricant's dilution ratio for HBK/FJ-2008.

4. Recommended nozzle

Specification	WA-100-10
Manufacturer	CY Spraying & Purification Technology
Nozzle/orifice	1.0 mm
Type	Automatic pressure-feed spray gun / air-liquid atomization
Published liquid output	~190 ml/min
Published spray width	~250 mm at 200 mm distance
Published air pressure	~3–4 bar
Published air consumption	~270 L/min
Connections	1/4 in. air and fluid
Approx. weight	~524 g

5. Initial operating trial

Parameter	Starting point
Nozzle	WA-100-10, 1.0 mm
Distance	~200 mm
Spray duration	~0.3–0.5 s; start near 0.5 s if coverage is insufficient
Liquid pressure	~0.15–0.25 MPa as a general trial range only if the nozzle can operate correctly there

Atomizing air	For WA-100-10, follow supplier's published 3–4 bar range
Pattern	Flat/wide fan; uniform thin coating
Adjustment	Tune spray duration and liquid flow based on coverage, release, die temperature and lubricant consumpti

At 190 ml/min, 0.5 s corresponds to about 1.6 ml of diluted lubricant. This is only a calculation from the published flow figure and should not be treated as a required production dose.

6. Other suppliers/equipment researched

Supplier/brand	Relevance
AYANK FORGING	Automatic die lubrication/cooling system; agitated tank, spray head and PLC/press synchronization.
Jinan Linteng	Automatic forging graphite spray equipment; confirm HBK/FJ-2008 compatibility.
Nanjing Descaling Technology	Automatic graphite spray station for forging lines; graphite-oriented.
Xuzhou Dayi Heavy Forging	Forging equipment and automation including die spraying/lubrication.
Steel Plant Specialities	Die-lubricant spray systems using tank/pump/agitation/spray-gun architecture.
BETE	Industrial lubrication and air-atomizing/flat-fan nozzle technology.
MAFA	TMKDUO release-agent spray gun; higher-cost European alternative.

7. Supplier links

Qingdao Xinyuan Forging Materials: https://www.qdxyfsm.com/

CY WA-100-10 / WA-101: https://cycospray.en.made-in-china.com/product/rwMtmUibnohF/China-Wa-101-Automatic-Spray-Tools-Assembly-Line-Special-Spray-Gun-for-Reciprocating-Machine.html

CY manufacturer page: https://cycospray.com/product/high-quality-hot-sale-pressure-feed-wa-101-wa200-new-pneumatic-automatic-paint-spray-gun/

AYANK FORGING: https://www.ayankforging.com/automated-mold-lubrication-and-cooling-equipment.html

Steel Plant Specialities: https://steelplantspecialities.com/die-lubricant-spraying-systems/

BETE lubrication: https://bete.com/application/lubrication/

MAFA TMKDUO: https://www.mafa-shop.com/en/Releasing-Agent-Spray-Gun-TMKDUO/180204-MAIN

8. Purchase recommendation

For initial testing, purchase two WA-100-10 units. Ask the supplier to confirm compatibility with HBK/FJ-2008 diluted 1:7, provide actual liquid flow at the intended pressure, and provide a spray-pattern test at 150–200 mm. Confirm wetted-material compatibility with the lubricant and water chemistry before production use.

9. Important limitations

The public sources reviewed did not establish a manufacturer-certified HBK/FJ-2008 dilution ratio or a nozzle setting specifically validated for the exact 600-t / 1200°C / Ø250–300 mm process. Final settings must be established by controlled trials using die coverage, release performance, die temperature and lubricant consumption as acceptance criteria. Supplier prices should also be reconfirmed before purchase.